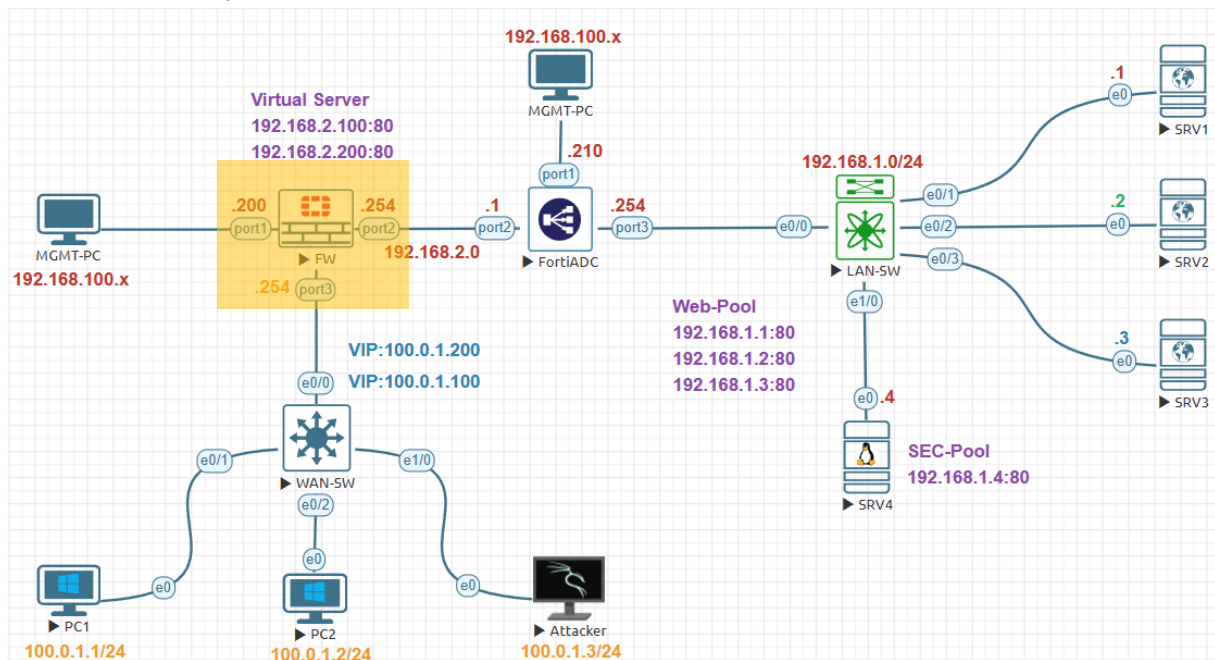


## Firewall Setup Lab:



|                                |                  |
|--------------------------------|------------------|
| Management Subnet              | 192.168.100.0/24 |
| FortiGate Management IP        | 192.168.100.200  |
| FortiADC Management IP         | 192.168.100.210  |
| Internal Servers Subnet        | 192.168.1.0/24   |
| FortiADC External Subnet       | 192.168.2.0/24   |
| FortiGate Firewall External    | 100.0.1.0/24     |
| FortiADC Internal IP Address   | 192.168.1.254    |
| FortiADC External IP Address   | 192.168.2.1      |
| FortiGate Firewall Internal IP | 192.168.2.254    |
| FortiGate Firewall External IP | 100.0.1.254      |
| SRV1 IP Address                | 192.168.1.1      |
| SRV2 IP Address                | 192.168.1.2      |
| SRV3 IP Address                | 192.168.1.3      |
| SRV3 IP Address                | 192.168.1.4      |
| Virtual Sever-1                | 192.168.2.100    |
| Virtual Sever-2                | 192.168.2.200    |
| FortiGate VIP-1                | 100.0.1.100      |
| FortiGate VIP-2                | 100.0.1.200      |
| External PC1 IP Address        | 100.0.1.1        |
| External PC2 IP Address        | 100.0.1.2        |
| External PC3 IP Address        | 100.0.1.3        |

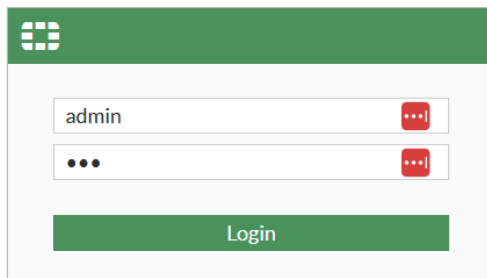
### Hostname Configuration

```
FortiGate-VM64-KVM # config system global
FortiGate-VM64-KVM (global) # set hostname FW
FortiGate-VM64-KVM (global) # end
```

### Management Interface Configuration

```
FW # config system interface
FW (interface) # edit port1
FW (port1) # set allowaccess https http ping ssh
FW (port1) # set mode static
FW (port1) # set ip 192.168.100.200/24
FW (port1) # end
```

⚠ Not secure 192.168.100.200/login?redir=%2F



A login form with a green header bar containing a white grid icon. Below the header, there are two input fields: the first contains the text 'admin' and the second contains three dots. Both fields have a red 'show/hide' icon to their right. Below the input fields is a green button with the text 'Login'.

### FortiGate Setup

⚠ Perform the following steps to complete the setup of this FortiGate.

- Specify Hostname ✓
- Change Your Password ✓
- Dashboard Setup
- Upgrade Firmware ✓

Begin

Later

### Setup Progress

Specify Hostname ✓

Change Your Password ✓

➤ Dashboard Setup

Upgrade Firmware ✓

### Dashboard Setup

**⚠** Select one of the following options to decide what dashboards will be available by default. You can always change your selection or manually customize your own dashboards later.

☒ **Optimal**  
A set of popular default dashboards and FortiView monitors.

☐ **Comprehensive**  
A set of default dashboards as well as all monitors and FortiViews. This set will be familiar to users coming from previous FortiOS versions

**OK** Later

FW

Dashboard

Status

Security

Network

Users & Devices

+

FortiView Sources

FortiView Destinations

FortiView Applications

FortiView Web Sites

FortiView Policies

FortiView Sessions

+

Network

Policy & Objects

Security Profiles

VPN

User & Authentication

System

Security Fabric

+

Add Widget

System Information

HostnameFW

Serial NumberFGVMEVS3DZRGUG18

Firmwarev7.0.9 build0444 (Mature)

ModeNAT

System Time2024/11/29 11:46:05

Uptime00:00:06:39

WAN IPUnknown

Licenses

FortiCare Support

Firmware & General Updates

IPS

AntiVirus

Web Filtering

FortiToken0 / 0

Unable to connect to FortiGuard servers.

Security Fabric

FW

Administrators

1 Console

1 HTTP

0 FortiExplorer

admin

super\_admin

Dashboard
Network
**Interfaces**
DNS
Packet Capture
SD-WAN
Static Routes
Policy Routes
RIP
OSPF
BGP
Routing Objects
Multicast
Policy & Objects
Security Profiles
VPN

FortiGate VM64-KVM

1 3 5 7 9 11 13 15 17 19 21 23

2 4 6 8 10 12 14 16 18 20 22 24

Create New Edit Delete Integrate Interface

Search

| Name                 | Type               | Members | IP/Netmask                    | Administrative Access            |
|----------------------|--------------------|---------|-------------------------------|----------------------------------|
| 802.3ad Aggregate 1  |                    |         |                               |                                  |
| fortilink            | 802.3ad Aggregate  |         | Dedicated to FortiSwitch      | PING<br>Security Fabric Connecti |
| Physical Interface 4 |                    |         |                               |                                  |
| port1                | Physical Interface |         | 192.168.100.200/255.255.255.0 | PING<br>HTTPS<br>SSH<br>HTTP     |
| port2                | Physical Interface |         | 0.0.0.0/0.0.0.0               |                                  |
| port3                | Physical Interface |         | 0.0.0.0/0.0.0.0               |                                  |
| port4                | Physical Interface |         | 0.0.0.0/0.0.0.0               |                                  |

FW

Dashboard
Network
**Interfaces**
DNS
Packet Capture
SD-WAN
Static Routes
Policy Routes
RIP
OSPF
BGP

Edit Interface

Name port1

Alias MGMT

Type Physical Interface

VRF ID 0

Role Undefined

Dedicated Management Port

Address

Addressing mode Manual DHCP Auto-managed by IPAM One-Arm Sniffer

IP/Netmask 192.168.100.200/255.255.255.0

FW
Dashboard
Network
Interfaces
DNS
Packet Capture
SD-WAN
Static Routes
Policy Routes
RIP
OSPF
BGP
Routing Objects
Multicast
Policy & Objects
Security Profiles
VPN

Edit Interface
Name: port2
Alias: FortiADC-LAN
Type: Physical Interface
VRF ID: 0
Role: LAN
Dedicated Management Port:
Address:
Addressing mode: Manual
IP/Netmask: 192.168.2.254/24
Create address object matching subnet:
Secondary IP address:
Administrative Access:
IPv4:
HTTPS:
PING:
FMG-Access:

Dashboard
Network
Interfaces
DNS
Packet Capture
SD-WAN
Static Routes
Policy Routes
RIP
OSPF
BGP
Routing Objects
Multicast
Policy & Objects
Security Profiles
VPN
User & Authentication
System

Edit Interface
Name: port3
Alias: WAN
Type: Physical Interface
VRF ID: 0
Role: WAN
Estimated bandwidth: 0 kbps Upstream, 0 kbps Downstream
Dedicated Management Port:
Address:
Addressing mode: Manual
IP/Netmask: 100.0.1.254/24
Secondary IP address:
Administrative Access:
IPv4:
HTTPS:
SSH:
PING:
SNMP:
FMG-Access:
FTM:

| + Create New ▾       |                    | Edit      | Delete                        | Integrate Interface                | Search |  |
|----------------------|--------------------|-----------|-------------------------------|------------------------------------|--------|--|
| Name ▾               | Type ▾             | Members ▾ | IP/Netmask ▾                  | Administrative Access ▾            |        |  |
| 802.3ad Aggregate 1  |                    |           |                               |                                    |        |  |
| fortilink            | 802.3ad Aggregate  |           | Dedicated to FortiSwitch      | PING<br>Security Fabric Connection |        |  |
| Physical Interface 4 |                    |           |                               |                                    |        |  |
| FortiADC-LAN (port2) | Physical Interface |           | 192.168.2.254/255.255.255.0   | PING                               |        |  |
| MGMT (port1)         | Physical Interface |           | 192.168.100.200/255.255.255.0 | PING<br>HTTPS<br>SSH<br>HTTP       |        |  |
| port4                | Physical Interface |           | 0.0.0.0/0.0.0.0               |                                    |        |  |
| WAN (port3)          | Physical Interface |           | 100.0.1.254/255.255.255.0     | PING                               |        |  |

Dashboard
Network
Interfaces
DNS
Packet Capture
SD-WAN
Static Routes
Policy Routes
RIP

### DNS Settings

DNS servers

Use FortiGuard Servers
Specify

Primary DNS server
8.8.8.8

Secondary DNS server
1.1.1.1

Local domain name

+

DNS Protocols

DNS (UDP/53)

i

Dashboard
Network
Interfaces
DNS
Packet Capture
SD-WAN
Static Routes
Policy Routes
RIP
OSPF
BGP

### New Static Route

Destination

Subnet
Internet Service

0.0.0.0/0.0.0.0

Gateway Address
192.168.100.1

Interface

MGMT (port1)

x

+

Administrative Distance
10

Comments

Write a comment...
0/255

Status

Enabled

Disabled

+ Advanced Options

Dashboard
Network
Interfaces
DNS
Packet Capture
SD-WAN
Static Routes
Policy Routes
RIP

### New Static Route

Destination
Subnet
Internet Service
192.168.2.0/24

Gateway Address
192.168.2.1

Interface
FortiADC-LAN (port2)

Administrative Distance
10

Comments
Write a comment...
0/255

Status
Enabled
Disabled

Dashboard
Network
Policy & Objects
Firewall Policy
IPv4 DoS Policy
Addresses
Internet Service Database
Services
Schedules
Virtual IPs
IP Pools

### New Virtual IP

VIP type
IPv4

Name
WEB-VIP-01

Comments
Write a comment...
0/255

Color
Change

Network

Interface
any

Type
Static NAT
FQDN

External IP address/range
100.0.1.100

Map to
IPv4 address/range
192.168.2.100

Dashboard
Network
Policy & Objects
Firewall Policy
IPv4 DoS Policy
Addresses
Internet Service Database
Services
Schedules
Virtual IPs
IP Pools
Protocol Options

### New Virtual IP

VIP type
IPv4

Name
WEB-VIP-02

Comments
Write a comment...
0/255

Color
Change

Network

Interface
any

Type
Static NAT
FQDN

External IP address/range
100.0.1.200

Map to
IPv4 address/range
192.168.2.200

|                  |   |                              |                             |                              |                        |        |
|------------------|---|------------------------------|-----------------------------|------------------------------|------------------------|--------|
| Dashboard        | > | <a href="#">+ Create New</a> | <a href="#">Edit</a>        | <a href="#">Clone</a>        | <a href="#">Delete</a> | Search |
| Network          | > | Name                         | Details                     | Interfaces                   | Services               |        |
| Policy & Objects | > | IPv4 Virtual IP 2            |                             |                              |                        |        |
| Firewall Policy  |   | WEB-VIP-01                   | 100.0.1.100 → 192.168.2.100 | <input type="checkbox"/> any |                        |        |
| IPv4 DoS Policy  |   | WEB-VIP-02                   | 100.0.1.200 → 192.168.2.200 | <input type="checkbox"/> any |                        |        |
| Addresses        |   |                              |                             |                              |                        |        |

|                           |     |             |  |
|---------------------------|-----|-------------|--|
| Dashboard                 | >   | Edit Policy |  |
| Network                   | >   |             |  |
| Policy & Objects          | >   |             |  |
| Firewall Policy           | ☆   |             |  |
| IPv4 DoS Policy           |     |             |  |
| Addresses                 |     |             |  |
| Internet Service Database |     |             |  |
| Services                  |     |             |  |
| Schedules                 |     |             |  |
| Virtual IPs               |     |             |  |
| IP Pools                  |     |             |  |
| Protocol Options          |     |             |  |
| Traffic Shaping           |     |             |  |
| Security Profiles         | >   |             |  |
| VPN                       | >   |             |  |
| User & Authentication     | >   |             |  |
| System                    | 1 > |             |  |
| Security Fabric           | >   |             |  |
| Log & Report              | >   |             |  |

Name **FortiADC-to-Internet**

Incoming Interface **FortiADC-LAN (port2)**

Outgoing Interface **MGMT (port1)**

Source **all** **+**

Destination **all** **+**

Schedule **always**

Service **ALL** **+**

Action **ACCEPT** **DENY**

Inspection Mode **Flow-based** Proxy-based

Firewall / Network Options

**NAT** **ON**

IP Pool Configuration **Use Outgoing Interface Address** Use Dynamic IP Pool

Preserve Source Port **OFF**

Protocol Options **default**



Dashboard

Network

Policy & Objects

Firewall Policy

IPv4 DoS Policy

Addresses

Internet Service Database

Services

Schedules

Virtual IPs

IP Pools

Protocol Options

Traffic Shaping

Security Profiles

VPN

User & Authentication

Edit Policy

Name

WAN-to-FortiADC

Incoming Interface

WAN (port3)

Outgoing Interface

FortiADC-LAN (port2)

Source

all

Destination

WEB-VIP-01

WEB-VIP-02

Schedule

always

Service

ALL

Action

ACCEPT DENY

Inspection Mode

Flow-based Proxy-based

Firewall / Network Options

NAT

Create New

Edit

Delete

Policy Lookup

Search

Q

Export

In

| Name                 | From                                                                                                     | To                                                                                                       | Source                                                                                  | Destination                                                                                                                                                                                      | Schedule                                                                                     | Service                                                                                   | Action                                                                                       | NAT                                                                                            |
|----------------------|----------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| FortiADC-to-Internet |  FortiADC-LAN (port2) |  MGMT (port1)         |  all |  all                                                                                                          |  always |  ALL |  ACCEPT |  Enabled  |
| WAN-to-FortiADC      |  WAN (port3)          |  FortiADC-LAN (port2) |  all |  WEB-VIP-01<br> WEB-VIP-02 |  always |  ALL |  ACCEPT |  Disabled |
| Implicit Deny        | <input type="checkbox"/> any                                                                             | <input type="checkbox"/> any                                                                             |  all |  all                                                                                                          |  always |  ALL |  DENY   |                                                                                                |